

Medical Software and Hardware Compliance Summary

I. Overall System Context

Development of a Rust-based hardware server for X-ray diagnostic control, communicating with a Python orchestrator and web UI. System classified as IEC 62304 Class B. Must comply with FDA, EU MDR, and IEC standards.

II. Core Software Architecture and Safety Principles

Rust server enforces physical safety via interlocks and watchdogs; orchestrator handles workflow and user interface. Communication via gRPC with mTLS. Safety logic includes enable switches, exposure limits, and signed configurations.

III. Roles in Requirements Definition and Validation

Requirements are defined by a multidisciplinary team: product owner (user needs), system engineer (system requirements), software architect (software requirements), hardware engineer (interlocks), QA/RA (traceability), cybersecurity lead (security), and human-factors specialist (usability).

IV. RACI Matrix Summary

Responsibilities distributed across team members for user needs, system requirements, risk management, coding, verification, and validation.

V. Purpose of Requirements Management Tools

Ensure every requirement is captured, linked to risks, implemented in code, verified by tests, and traceable for FDA audits. Tools: Jama, Polarion, Codebeamer, Doorstop, GitLab Issues, ReqView.

VI. Open-Source Toolchain for Medical Startups

Suggested stack: GitLab CE/Jenkins (CI/CD), Doorstop (requirements), pytest + Allure + Kiwi TCMS (testing), SonarQube/Clippy/cargo-audit (analysis), Syft/CycloneDX (SBOM), Markdown + Pandoc (documentation).

VII. Documentation & Traceability Structure

Organized Design History File (DHF) with plans, requirements, design, code, verification, release, and maintenance folders for full audit trail.

VIII. Essential Standards

Software: IEC 62304, AAMI TIR45. Hardware: IEC 60601-1 family. Cross-cutting: ISO 13485, ISO 14971, IEC 62366-1, FDA Cybersecurity, ISO 27001, FDA 21 CFR 11/820.

IX. Standards Interconnection Map

ISO 13485 (QMS) → ISO 14971 (Risk) → IEC 62304 (Software), IEC 60601-1 (Hardware), IEC 62366-1 (Usability), FDA Cybersecurity (2023), ISO 27001 (Security).

X. Estimated Cost of Standards

Software-only: €800–1,000. Full system: €2,000–2,500. Extended set with cybersecurity: €3,000–3,500.

XI. Key Practical Insights

FDA requires traceability, not specific tools. Every requirement must link to a risk control, test, and code module. Logs, commits, and tests serve as regulatory evidence.

XII. Implementation Roadmap

Phase 1: Core standards and toolchain. Phase 2: Add 60601 hardware standards. Phase 3: DHF preparation, cybersecurity verification, validation testing, design reviews.